

Tangible E-Textile Interactive Interface for Digital Patternmaking

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ABSTRACT

A common method of generating a garment pattern is by draping fabric on a 3D form. Many apparel designers prefer this tangible process rather than the more abstract process of drafting patterns in 2D using CAD software, but the visualization benefits of 3D rendering remain important to product development. Currently, draped patterns must be manually digitized to transfer the shape to the CAD environment, a cumbersome process. Here, we explore augmenting the fabric used to drape patterns with e-textile components, such that pinning the fabric to the form directly digitizes the pattern shape. The e-textile interface uses a keypad matrix approach with rows and columns on opposite sides of the fabric. Passing a metallic pin through a row/column intersection creates a circuit connection that a microcontroller reads and maps to a 2D spatial layout on the screen. The proof-of-concept device developed here is a first step toward an automatic digitizing system that allows designers to preserve manual skills and processes while integrating more efficiently with digital systems.

CCS CONCEPTS

• Human-Computer Interaction; • Electronic Textiles; • Digital Patternmaking;

KEYWORDS

E-textile interface, Patternmaking, Keypad matrix, Draping

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1 INTRODUCTION

Tangible HCI has brought computational functionality to everyday objects. In [1]–[3], embedded electronic capabilities serve to render an existing tangible tool or object responsive to a user's interactions. For textiles, this is a slightly more complex challenge, as the introduction of electronics often changes the mechanics and texture of fabrics. In apparel design, fabric is both a tool and a

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Figure 1: - E-Textile interface draped on a half-scale mannequin.

material – in manual patternmaking processes, fabric is draped on a 3D form to derive the shape of the garment, which then becomes the garment pattern. The designer places a piece of fabric (either the same fabric that will be used in the target garment, or a neutral placeholder fabric like plain-weave cotton muslin or toile fabric) on the mannequin, with attention to the orientation of the grain of the fabric. The fabric is pinned to the mannequin in strategic places, both to secure the fabric and to mark edges and contours. The fabric is folded, cut, stretched, and pinned into the shape of the garment. Landmarks are then captured by marking on the surface of the fabric. Once the fabric is removed, these landmarks are corrected (trued) and connected into the outline of the piece shape. This shape can then be transferred to pattern paper or into a digital system, where seam allowances can be added. In a digital system, a virtual sample of the garment can be generated, or the piece can be printed and used to cut garment pieces out of the final fabric. Physical pattern pieces are often digitized for storage, so that they can be more easily edited during the fitting/product development process or modified into new shapes. Digital patterns are essential for downstream manufacturing steps like creating a size range and developing cutting layouts.

Here, we explored augmenting the textile itself to create a tangible input tool for patternmaking. Balancing functionality with



Figure 2: – Digitized piece contour plotted on a processing software graphical interface.

the inherent properties of textiles comes with its own challenges. Preserving the mechanical properties of textiles is important, so textile- and fiber-based electronic components are essential [4], [5]. Similarly, preserving the tools and processes traditionally used by designers is also important. Here, we developed a fabric-based input device that uses standard sewing pins to create circuit connections and capture the geometry of the pattern shape.

2 SYSTEM DESIGN AND CHARACTERIZATION

A proof-of-concept device (Figure 1) was fabricated by forming a "keypad matrix" circuit with silver coated Nylon conductive thread traces (Shieldex) stitched on two sides of a woven fabric using the Brother BAS-342G pattern stitching machine (although any type of lockstitch sewing machine can be used) (Figure 3). The matrix's rows and columns were connected to input and output pins, respectively, of an Arduino microcontroller. Passing a metal pin through the fabric at a row/column intersection completes a circuit, and the input is read by the microcontroller. The keypad matrix approach allows the microcontroller to know which row and column are active and, therefore, the x/y position of the pin (Figure 5). The input is then logged as a point on the digitized piece's contour.

In the prototype interaction process, a designer would drape using the e-textile matrix on a mannequin, pinning the fabric outside of matrix intersection points, much as they would normally with standard fabric. When the desired shape is achieved, instead of marking contour points with a pencil, the designer would place a pin through the matrix intersection closest to the desired point, and then remove the pin. The input is then logged as a point on the digitized piece's contour on processing software graphical interface (Figure 2). Contour points are input sequentially in a clockwise or counterclockwise fashion, and a closed shape is built from those input points on the screen. The shape can be saved to a standard .pdf format when the input is complete and imported into common CAD software packages.

The prototype interface was developed in half scale, for simplicity and conservation of materials. However, an interface like this is



Figure 3: - Stitching the matrix using Brother BAS-342G pattern-stitching machine.

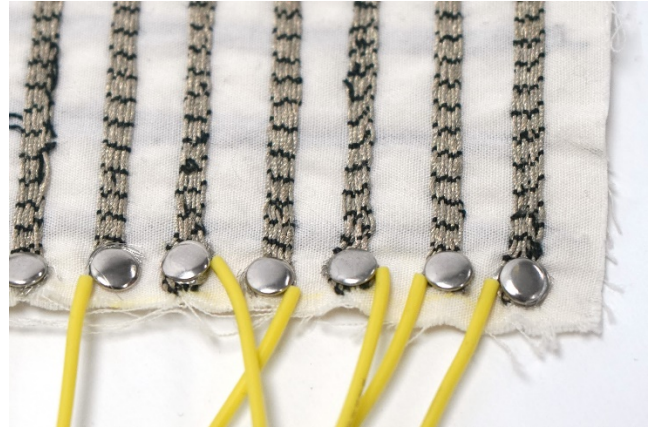


Figure 4: Trace width was increased by stitching the traces six times.

conductive to easy translation between half-scale development and full-scale digital patterns. Half-scale mannequins are used for many reasons in apparel product development, including conservation of materials, ease of manipulation of smaller pieces, and feasibility of prototyping on a wider variety of body type [6],[7]. We checked the electrical performance of the stitched traces by checking for continuity and measuring the electrical resistance values for each row and column using the digital multimeter. The electrical resistance for unconnected traces (without the pin) ($M=0.83\Omega$, $SD=0.087\Omega$) and for connected traces (with the pin at the row-column intersection) ($M=1.78\Omega$, $SD=0.35\Omega$). The electrical resistance of the augmented fabric is relatively low, which is desirable for e-textile applications.

3 DISCUSSION

A key challenge in forming the rows and columns of the matrix was balancing the thread machine tension to avoid short circuits between the two sides of the fabric. The system relies on rows

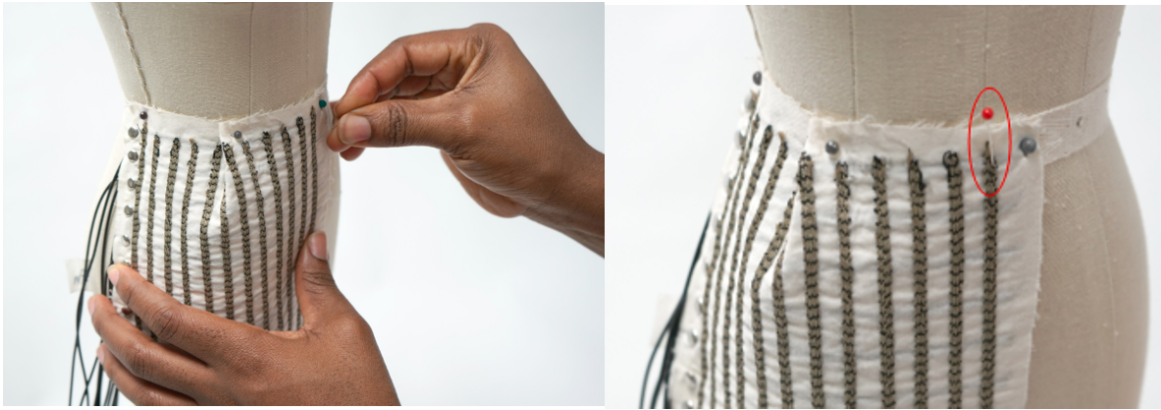


Figure 5: - (left) Pinning the x/y position and (right) pinned x/y location (in red) on the textile matrix.

and columns being isolated by the fabric substrate, and we use an unbalanced lockstitch to keep the conductive thread on one side only. For thicker fabric and smaller matrices (up to 6x6), a 4mm stitch length was successful. For larger matrices and thinner fabric, it was necessary to use two pieces of fabric joined with a fusible adhesive. To improve the ease of locating row/column intersections, we increased the width of the traces by stitching each row and column six times (Figure 4).

Because of the masking effect that keypad matrices experience with simultaneous inputs, we adopted a serial-input paradigm where only one pin connection is made at a time. The diode-based approach commonly used to circumvent masking is not feasible in our textile layup, as diodes must be connected between the rows and columns during input. While the proof-of-concept prototype demonstrates that it is possible to use such an e-textile matrix to capture draped pattern shapes, there are several areas in which this system requires further development to adequately support the full range of draping practice. In its current implementation, the matrix is limited in resolution (as shown in the relatively crude pattern shape shown in Figure 1). That is largely driven by the limitations on input/output pins for the microcontroller used here and can be avoided using a different hardware setup or by using more advanced multiplexing. The effect of a higher-resolution matrix on cabling and trailing wires should also be considered, as it affects the usability and drape of the fabric (as seen in Figure 6).

The conductive rows/columns here also affect the mechanics of the fabric. There are many ways in which conductive traces can be integrated into a textile (weaving/knitting, printing, stitching), and these can be further investigated to determine optimal integration methods for different textiles. Currently, the prototype only supports draping of woven textiles, as the matrix does not stretch. To facilitate draping in stretch fabrics, the matrix would need to be fabricated using stretchable traces. The fabrication of the current prototype also poses challenges for interaction in the draping process. As the traces here are un-insulated, they can create short circuits when the fabric is folded (such as during formation of darts or pleats). Insulated conductors would not short circuit but would be more difficult to pierce with the pin. A surface insulation (film

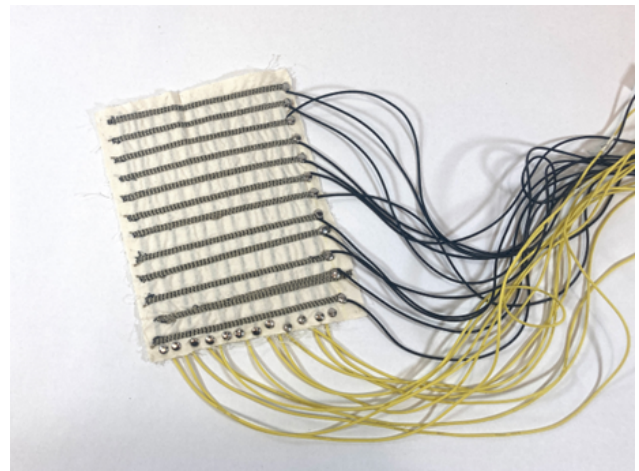


Figure 6: Matrix fabric swatch showing the rows and columns on the two fabric sides with trailing wires for connection.

or coating) may be more effective. In addition, the current implementation only supports cutting on two sides (the opposite sides from the integrated connectors). Draping practice often involves clipping into the fabric to create flare. In our prototype, it is possible to achieve that kind of process if the matrix is oriented appropriately, but very complex shapes that might require clipping from more than 2 sides would not be possible, and clipping actions would destroy the matrix. An implementation that uses more disposable materials may facilitate enough clipping for typical pattern shapes, if not for the most complex shapes. Finally, the current processing software only registers one type of point, and cannot distinguish curves, darts, or notches. These attributes can be modified after import in the CAD system, but the ability to directly record different point types during draping would improve usability.

4 CONCLUSION

This proof-of-concept device demonstrates that digitizing patterns during draping and using draped textile and pins directly is possible.

As an initial prototype, there are many areas of necessary improvement, particularly in the resolution and mechanical properties of the matrix, as well as insulation of the integrated traces. However, as an interaction paradigm it demonstrates the feasibility of a tangible interface for creating digital garment patterns, through practice rooted in patternmaking tradition.

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